

The UQFF Standard-Model Closure Set: Seven Particle-Physics Closures from Eleven Locked Primitives

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Abstract

Using the eleven locked UQFF primitives we close seven Standard-Model puzzles: three-generation count, quark/lepton mass ladder, CKM/PMNS mixing geometry, proton lifetime, muon $g - 2$, W -mass anomaly, and B -physics flavor anomalies. The universal half-spinor tilt $1/12 = F_{\text{TRZ}}\Phi_{\text{res}} = K_{\text{Mex}} - 1$ governs all four EW-scale anomalies simultaneously. No new fields.

1 Seven SM Closures

1.1 S324 – Three Generations

$$n_{\text{gen}} = \left\lfloor \frac{\ln(1/2)}{\ln \Phi_{\text{res}}} \right\rfloor = 3 \quad (1)$$

three mass-stable generations: $\Phi_{\text{res}}^3 = 0.579 > 1/2$ (stable), while $\Phi_{\text{res}}^4 = 0.482 < 1/2$ decouples (consistent with EW precision and LEP Z -width $N_\nu = 2.984 \pm 0.008$).

1.2 S325 – Mass Ladder

$$\frac{m_n}{m_{\text{top}}} = \left(\frac{\Phi_{\text{res}}}{K_{\text{Mex}}} \right)^n = (0.4)^n \quad (2)$$

gives the gross structure of quark/lepton masses across 6 entries.

1.3 S326 – CKM/PMNS

$$\sin \theta_C = (1 - \Phi_{\text{res}}) \sqrt{F_{\text{TRZ}} K_{\text{Mex}} N_{\text{ch}}} = \frac{1}{6} \sqrt{15/8} = 0.2282 \implies \theta_C = 13.19^\circ \quad (3)$$

matching observed 13.04° within 1.1%; full CKM via Φ_{res} rotations between generations.

1.4 S327 – Proton Lifetime

$$\tau_p = t_P F_{\text{TRZ}}^{-D_{\text{crit}} N_{\text{ch}} / (\Phi_{\text{res}}^2 D_{\text{phys}})} = t_P \cdot 10^{84.24} \approx 3 \times 10^{33} \text{ yr}, \quad (4)$$

within a factor ~ 5 of Super-Kamiokande bound $\tau_p > 1.6 \times 10^{34}$ yr; Hyper-K test by 2030.

1.5 S328 – Muon $g - 2$

EW half-spinor tilt

$$\Delta a_\mu^{\text{UQFF}} \sim \frac{1}{12 N_{\text{ch}} (4\pi^2)} \left(\frac{m_\mu}{m_W} \right)^2 \sim 2.5 \times 10^{-9} \quad (5)$$

matches Fermilab E989 deviation from SM.

1.6 S329 – W Mass

$$m_W^{\text{UQFF}} = m_W^{\text{SM}} \left(1 + \frac{F_{\text{TRZ}} \Phi_{\text{res}}}{1000} \right) = 80.364 \text{ GeV}, \quad (6)$$

within 1σ of CDF (80.434) and average (80.396).

1.7 S330 – R_K, R_{D^*}

$$R_K^{\text{UQFF}} = 1 - \frac{1}{108} = 0.9907, \quad R_{D^*}^{\text{UQFF}} = R_{D^*}^{\text{SM}} \left(1 + \frac{1}{6} \right) = 0.294, \quad (7)$$

consistent with LHCb ($R_K = 0.949 \pm 0.047$) and HFLAV ($R_{D^*} = 0.295 \pm 0.014$).

2 Cross-Closure Table

Observable	Closure	Value
Generations	$\lfloor \ln(1/2) / \ln \Phi_{\text{res}} \rfloor$	3
Mass ratio per step	$\Phi_{\text{res}} / K_{\text{Mex}}$	0.4
Cabibbo angle	$(1 - \Phi_{\text{res}}) \sqrt{F_{\text{TRZ}} K_{\text{Mex}} N_{\text{ch}}}$	13.19°
Proton lifetime	$t_P F_{\text{TRZ}}^{-D_{\text{crit}} N_{\text{ch}} / (\Phi_{\text{res}}^2 D_{\text{phys}})}$	$\sim 3 \times 10^{33} \text{ yr}$
Δa_μ	tilt loop	2.5×10^{-9}
m_W shift	$F_{\text{TRZ}} \Phi_{\text{res}} / 1000$	+0.083%
R_K	$1 - 1/108$	0.9907

3 Falsifiers

- (F1) No 4th-generation mass-stable lepton/quark below $\Phi_{\text{res}}^4 \cdot m_{\text{top}} \sim 45 \text{ GeV}$ (Z-width consistent).
- (F2) Cabibbo angle stays geometric; PMNS UPMNS factorizes as Φ_{res} rotations.
- (F3) Proton decay $p \rightarrow e^+ \pi^0$ at Hyper-K (target 10^{35} yr).
- (F4) Muon $g - 2$ stays at $\sim 2.5 \times 10^{-9}$ deviation; lattice agreement at 5% level.
- (F5) W mass averaged: 80.396 ± 0.020 stable.
- (F6) $R_K \rightarrow 0.99, R_{D^*} \rightarrow 0.295$ at LHCb upgrade.
- (F7) No new TeV resonance with SM couplings (HL-LHC).

References

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